PSAI_101

SPIDERDAO GRANT PROPOSAL

- **Proposer:** [SpiderDAO] (https://github.com/SpiderDAO)
- **Payment Address:** 3Pxq3ViqRW6b3e7qsX7Mo7gSikHcToa7ig
- **Document Version:** Version 2.4
- **Status:** Terminated

PROJECT OVERVIEW :PAGE_FACING_UP:

SpiderDAO is a next-generation hardware-based DAO governance model that aims to bring a new standard of fairness to existing DAO frameworks. Traditional DAO models are subject to attack and takeover by plutocracies controlled by wealthy whales. He who controls the votes controls the DAO.

OVERVIEW

SpiderDAO is a unique governance infrastructure layer for a hardware-enabled DAO and can be applied in a variety of use cases. As the first use case, Spider will leverage its well-established presence in the hardware VPN market together with a partnership with Sentinel, an established dVPN provider and BPSAA Alliance member, to create a fully self-governing decentralized VPN network called SpiderVPN.

We intend for SpiderDAO to run on its own Parachain on the Polkadot network. If the bond cannot be funded, the protocol will run on Parathreads or possibly as a series of smart contracts. Our initial testnet and protocol configuration will be built on Kusama and ported onto Polkadot for the mainnet launch.

We believe that in order to fulfil the SpiderDAO’s vision of bringing greater privacy to the end-user, it is vital to establish a robust ecosystem that prioritises users’ interests. To do this, we must first establish a resilient, scalable and community-led governance system that dynamically adapts to the rapidly changing times. We believe that building a decentralised autonomous organisation is the perfect mechanism that will enable us to reach this goal.

PROJECT DETAILS

SpiderDAO’s long term vision is to propose a set of tools bringing privacy and security to today and tomorrow’s internet users. To do so in an efficient and sustainable fashion, SpiderDAO’s team imagined and leveraged a robust DAO standard that will withstand the incredible adversity that such a mission implies. A first step is to address the plutocracy governance problem in traditional DAOs by introducing a set of rules that must be abided by to qualify for the right to an on-chain vote via the DAO.

By deploying a combination of hardware and software tools on top of the Polkadot Governance protocol, SpiderDAO bakes-in an inherently democratic, whale-resistant governance mechanism. This enables the evolution of the network, feature development decisions and the subsequent value creation to be distributed fairly amongst its users instead of being manipulated by a well-resourced, centralised party.

This is achieved through the introduction of a unique governance model with three components - SpiderConnect Hardware Router, the SPDR token and the Spider Virtual Private Network. The SpiderDAO utilises this dual-speed model to balance the voting power away from DAO token holders (as is in traditional DAOs) and towards the hardware owners. The SpiderConnect Hardware Router can be purchased by any community member, once distributed, the owner of the router can then connect it to the Spider Virtual Private Network via the Spider Dashboard.

SpiderDAO ecosystem brings together several other components to provide the building blocks for growing and sustaining a dynamic ecosystem.

Spider Ecosystem will include:

- SpiderDAO is the first hardware-governed DAO with a self-maintaining and self-improving robust Hardware/Software Governance control mechanism.
- SpiderConnect Routers offer multiple roles as a high-speed VPN tunnel, a DAO voting ticket, and a node in a fully autonomous decentralised VPN.
- SpiderVPN is a Virtual Private Network interlinking all the elements of the DAO & offering a complimentary dVPN service for qualifying users.
- SpiderToken (SPDR) - various use cases including; Eligibility for voting rights as a SpiderDAO participant, Liquidity Mining and Liquidity as Utility (LAU) and a substitute payment for SpiderVPN services.
- SpiderMarket will be an easy way to Sell & buy Bandwidth for SPDR rewards based on the Rules in the Service Level Agreement (SLA).
- SpiderVault which will be created to accumulate 2% of Crypto proceeds generated by services utilizing the SpiderDAO which can be redistributed for future propositions.
- **SpiderDashboard** (as seen Below) will provide an end-users access point to the features of the SpiderDAO.

ECOSYSTEM FIT

We see many projects in the Polkadot ecosystem that are having similar issues when it comes to voting system: MANTRA DAO is one of the examples.

The main problem with **Mantra DAO** which is built on **Rio Chain** is the same problem that many other DAOs faces where each token represent 1 vote in the ecosystem which introduces a major problem with the governing body where whales can take control of the DAO by buying big amounts of tokens making them in a way the deciding vote for any proposals.

TO SOLVE THIS WE INTEND TO ADD THE FOLLOWING FEATURES

SpiderConnect Router Security

- Our routers do have a secured ROM which we can control securely for all units and that's where the essential data will be saved. Our unique ID which currently contains the mac address & the router serial number will be combined with the wallet address and encrypted using our algorithm to make sure it would be nearly impossible to be compromised.

Whale Resistant Voting Model

- Our model shifts voting power away from token holders to hardware owners, where one validated router equals one vote. Additionally, Blockchain-based router tracking ensures that hoarding of devices are instantaneously detected by the network and 'fake' routers are locked out of the network. The liquidity incentive mechanism will be available to all users and not just people with routers so the entry barrier is removed in this case.

Voting by Direct Democracy

- Our voting mechanism would be classified as a Direct Democracy system. We are going to balance the voting power by only allowing token holders to have one router regardless of how many tokens they hold. We will make sure they got to go through SpiderConnect to gain more routers for their tokens if we agreed to that. Our DPI filters will monitor that keeping their identity anonymous.

How will spiderDAO try and prevent Whales/Bots from manipulating the Hardware Governnace?

- We are aware that whales/bots can buy more routers using new wallets to try and affect the governance model, but that will come at a heavy cost for adding more routers and also have to wait for a period of time deemed reasonable to the DAO. Users must have active transactions on the wallet before being able to vote to make sure bots and whales won't get access with ease.

- We will be keeping a log of every MAC Address and serial number for every router we add to our inventory, this we'll be added to an on-chain ledger or on-chain database. This will help us to identify any clone or anyone trying to spoof/manipulate the hardware. We will also be in constant contact with the growing SpiderDAO community and the open-source community for beneficial ideas on how to make the governorate safer and more resilient to whales-resistance.

- If the essential data (explained below) is not matched to the end-user and they try to spoof or manipulate access. There will be 3 warnings followed by a ban from taking a part in our governance model for a limited time(still to be decided). The ban time will be increased systematically based on the number of times this is identified until the customer is banned indefinitely if the infringement persistently continues.

What is the essential data?

The essential data is a combination of the following:

- Encrypted wallet seed using aes-256-cbc encryption standard with an auto-generated passphrase* saved as part of the essential data.
- Router Mac Address to avoid it being cloned.
- Router serial number which cannot be cloned as it's a part of the chipset code.
- SHA-512 checksum for a temp file that is created with all the data above which is updated every time the router is rebooted with the timestamp.

PS: Please note the passphrase would only be local and will be independent to the requests sent by the router for the governorate operations.

Is SpiderConnect Hardware Router open-source?

- Yes as we want to be involved much more in the open-source community, We are going to be using the Uboot environment with its open-source code to save the essential data which would be listed in the next question.

- To go into more detail on that point, We will also be giving our community members the option to choose between Open Source wireless drivers and the closed source ones for them to choose so we can make sure we offer an OpenSource friendly solution.

PS: No secured or closed source ROM will be needed using this solution

How we plan to build on Polkadot Ecosystem?

- We are planning to build using smart contracts at the beginning but as we scale up to more transactions, We are most likely to move toward a shared solution so we can cut the costs and take charge of how our blockchain will operate.

How we plan to achieve on-chain and off-chain communication with SpiderConnect Routers and SpiderDAO governance?

- We are planning to write a substrate faucet to communicate with our discord server for our customers to send the token needed for the vote, Initiate voting, pass on the vote for the voting process that would be operated through our discord server.
- Real votes will be done through our substrate faucet until the final Polkadot/SpiderDAO flavour features are fully-fledged.

TEAM :BUSTS_IN_SILHOUETTE:

TEAM MEMBERS

- Nathan Varty (Full-Time, Founder, CEO)
- Žiga Flis (Full-Time, Founder, Senior blockchain Developer & Analyst)
- Anas Sayed (Full-Time, Founder, CTO, Development Manager, Architecture, Full Network Stack Implementation)
- Alexy Petrunin (Full-Time, Senior Backend Developer)
- Dr. Alfie Zhao (Advisor, Hardware Design)
- Pierre Laurent (Strategic Advisor, Blockchain and crypto Enthusiast)

TEAM WEBSITE

- https://spiderdao.io

LEGAL STRUCTURE

SpiderDAO is incorporated in Cayman Island.

TEAM'S EXPERIENCE

Nathan Varty

- Experienced entrepreneur specialized in IoT development and operations, Skilled in management of onsite Developers.
- Successfully overseeing SpiderVPN's operations since 2017.
- Passionate about decentralizing global currency and governance with the vision to implement that in wider fields.
- Over 5 years experience investing and trading in digital assets.

Žiga Flis

- Worked as a Chief Technology Officer at RiveX, where he contributed to the development of various blockchain technologies.
- Experienced in on and off-chain tech and has a passion for developing decentralized solutions.
- Experienced in crypto markets since 2016.

Anas Sayed

- Long standing Project Manager and working with teams of Developers and Administrators.
- Worked on many projects in Entertainment/Telecommunications field including FuboTV, Pensil Media, SpiderVPN.
- Currently working on different projects with market leaders in Telecommunications field.
- Bachelor's Degree in Computer Science from HICIT in Shorouk Academy, Egypt.

Alexy Petrunin

- Experienced backend developers with full SDLC implementations.
- Developed multiple OpenWRT projects.
- Extensive knowledge in all Open Systems Interconnection (OSI) model.

Dr. Alfie Zhao

- Chief Technology Officer, GL Technologies (Hong Kong) Limited.
- Winner of CES innovation awards in 2019 and 2020.
- PhD in engineering management with proven track record in hardware IoT since 2012.

Pierre Laurent

- Co-founder at Atka advisory.
- Overseeing the European operation of a cryptocurrency exchange as Blockchain Partnerships Manager in 2018
- Early adopter of Blockchain and advisor on several projects
- Trading and investing in different projects since 2016
- Tech-savvy, startup enthusiast, passionate about blockchain technologies.

TEAM CODE REPOS

- https://github.com/spiderdao
- https://github.com/painfull30
- https://github.com/flisko

TEAM LINKEDIN PROFILES

- **Nathan Varty** https://www.linkedin.com/in/nathan-varty-b07235162/
- **Žiga Flis** https://www.linkedin.com/in/%C5%BEiga-flis-3112896b/
- **Anas Sayed** https://www.linkedin.com/in/anas-sayed-34a99b73/
- **Alexy Petrunin** https://www.linkedin.com/in/alexey-petrunin-aa06807a/
- **Dr. Alfie Zhao** https://www.linkedin.com/in/jianbin-zhao-2b04567/
- **Pierre Laurent** https://www.linkedin.com/in/pierrelaurent789/

DEVELOPMENT ROADMAP :NUT_AND_BOLT:

OVERVIEW

- **Total Estimated Duration:** 2 Months/POC
- **Full-time equivalent (FTE):** 15.57
- **Total Costs:** 1.67 BTC

MILESTONE 1 — SUMMARY

Number	Deliverable	Specification
0a.	License	Apache 2.0
0b.	Documentation	We will provide both inline documentation of the code and a full tutorial that explains how to interact and communicate with the testnet protocol
0c.	Testing Guide	The code will have proper unit-test coverage (>90%) to ensure functionality and robustness. In the guide we will describe how to run these tests
1.	Testing	We will conduct testing of the developed functionalities on Westend testnet.
2.	DAO framework	We are going to build our DAO framework using Ink! Smart Contracts for this milestone.
3.	Discord Communication	Write a substrate faucet to establish communication between SpiderConnect Hardware Router & discord server allowing contribution to SpiderDAO by allowing users to send the token needed for the vote, Initiate voting, pass on the vote for the voting process that would be operated through our discord server.
4.	Build TestNet	The testnet will be based on Substrate framework for this milestone.
5.	Test SpiderDAO	Successfully connect DAO entities (individual voters) to Spider routers. This will include one use case to show the connection is established correctly using Discord communications. We will be using Python with the help of https://github.com/polkascan/py-substrate-interface library.
6.	Medium Article	Create a medium article to announce all deliveries to the community once passed evaluation.
7.	Repository	Repository including a README that describes the milestone and explains how to run, test and contribute
8.	Docker	A docker container that will also run on CI to test the deliverables of the milestone

COMMUNITY ENGAGEMENT

We plan to write several pieces about SpiderDAO/SpiderVPN implementation. They include, but are not limited to:

- Weekly Medium Articles
- Daily Twitter Threads and Posts
- We also intend to engage community by running range of incentivised testnets to get more feedback from the existing end-users to improve our product.
- Educating an Ideal processes of hardware voting policy in a DAO structure

FUTURE PLANS

Future Developments will focus on the following:

- Incentive programs for validators
- Bug bounties
- Hackathon
- Grants to develop new products under the SpiderDAO Umbrella
- Decentralized SpiderChat | SpiderCloud | SpiderMail

Open Source Development Opportunities

- Hardware DAO Development
- Networking Routers
- Operating Systems

ADDITIONAL INFORMATION :HEAVY_PLUS_SIGN:

Any additional information that you think is relevant to this application that hasn’t already been included.

- **What work has been done so far?** SpiderVPN has been operating as a company for 3 years, with over 6000 customers and users of the router. A userbase to eventually be leverage for the future of the project. White paper and conceptualization are complete, high-level architecture has been designed, and development roadmap has been created. Actual development has begun.
- **Are there are any teams who have already contributed (financially) to the project?** Our founding team has covered all costs accrued thus far out-of-pocket. We have an investor on the sidelines who will cover any shortfall that may arise between the use of this grant and our next source of funding, whether that be the General Grants program or a VC-led private sale.
- **Have you applied for other grants so far?** No
- **How can I get involved?** Anyone looking to get involved is more than welcome to reach out to info@spiderdao.io Telegram groups and community creation/outreach will occur once the testnet is in use and we are preparing to launch.